

* SANDISK CEREBRUM - 2026

VITC VERIFICATION TRACK -

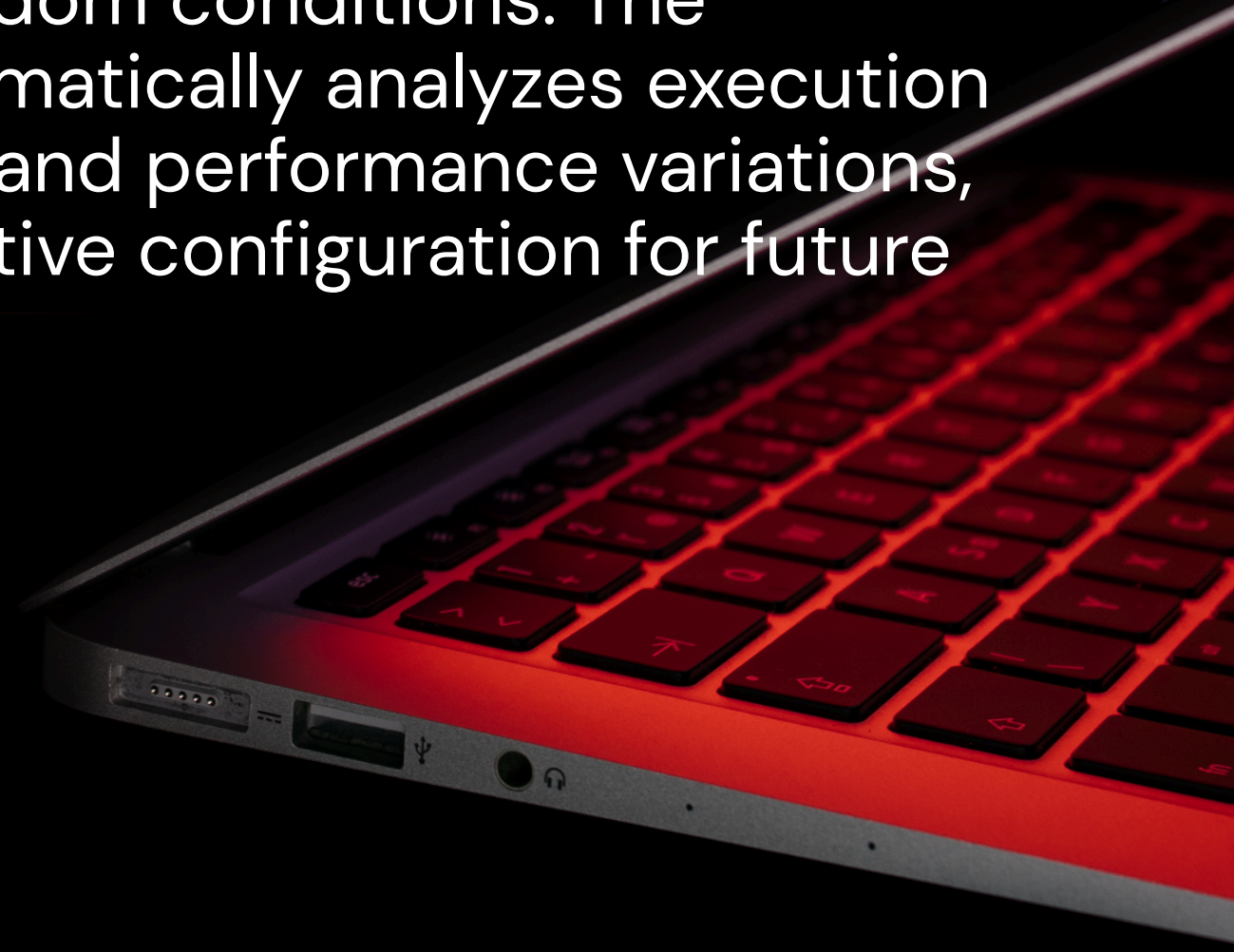
AI-POWERED CONFIGURATION INTELLIGENCE
FROM EXECUTION LOGS

▶ THE ENGINE OF MODERN
TECHNOLOGY

TEAM DETAILS: TEAM NANOMETER
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► PROBLEM STATEMENT

- Modern semiconductor and VLSI systems undergo thousands of test executions with different configurations, workloads, random seeds, and environmental conditions. When failures occur, engineers often have to manually analyze large volumes of logs to determine what caused the failure and which configuration led to it. For example, in a VLSI verification environment, a test may pass under one configuration but fail under another due to a specific combination of parameters or random conditions. The challenge is to develop an AI-powered dashboard that automatically analyzes execution logs and test data, identifies the key factors behind failures and performance variations, predicts potential failures, and recommends the most effective configuration for future test runs.

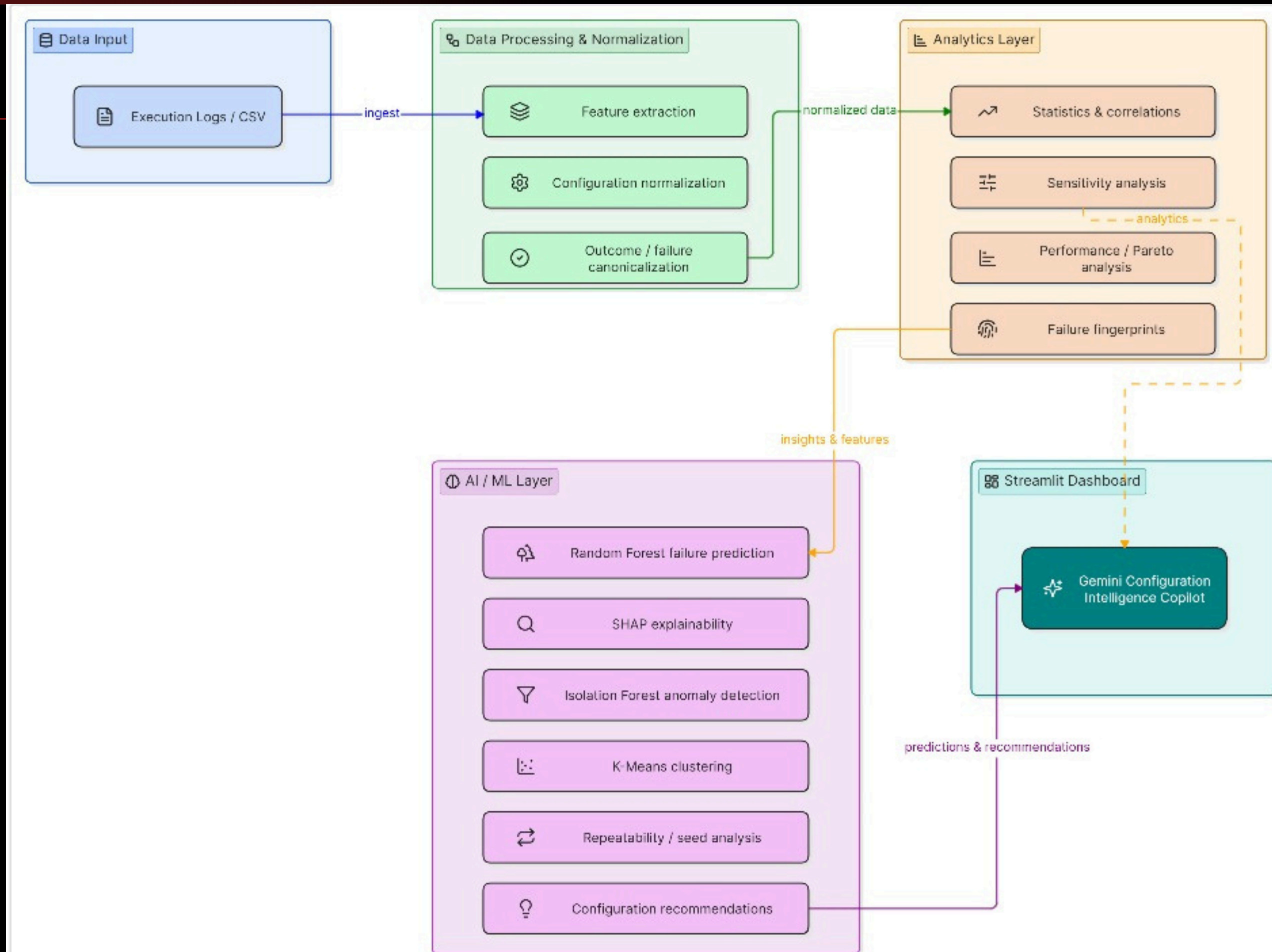


OUR SOLUTION: PILOT

We developed : PILOT (Performance & Intelligence from LOGs for Optimal Testing) :

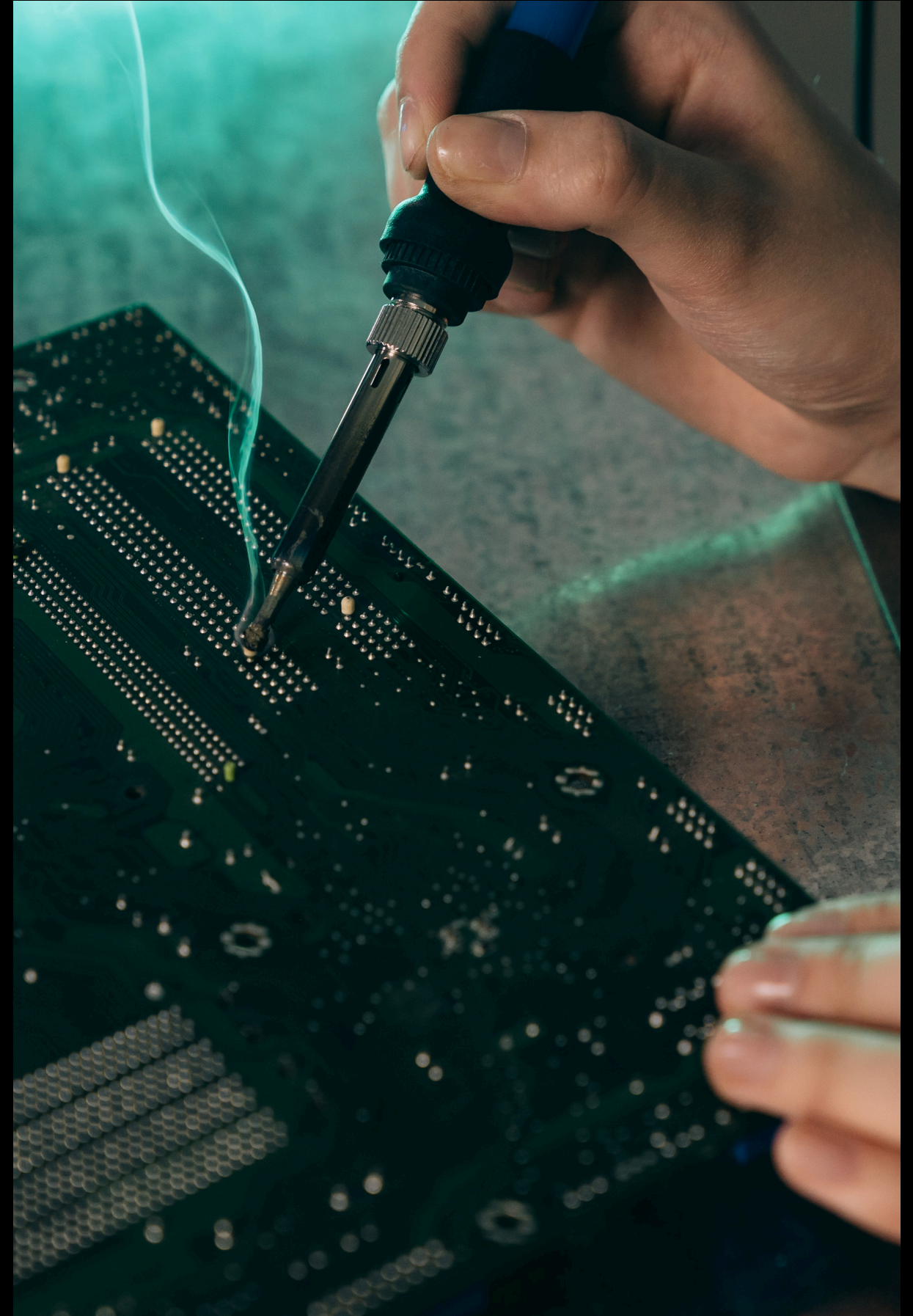
- An AI/ML-powered web dashboard that automates the analysis of large-scale VLSI and semiconductor execution logs.
- The platform accepts thousands of execution-log files/CSV data, processes and normalizes configurations, and generates interactive visualizations for performance, failures, correlations, sensitivity, anomalies, and configuration stability.
- Machine-learning models provide failure-risk prediction, anomaly detection, clustering, explainable insights and graphs, and optimized configuration recommendations for errors
- An in-built integrated Gemini Configuration Intelligence Copilot allows users to query the results in natural language and understand why failures occurred and what configuration should be used next.

ARCHITECTURAL FLOW:



INNOVATION:

- Unified AI + Analytics Platform: Combines log analysis, statistical analysis, ML models, and an AI Copilot in one dashboard.
- Automated Failure Analysis: Identifies failure patterns, fingerprints, anomalies, and recurring issues without manually inspecting thousands of logs.
- Explainable AI: Uses Random Forest + SHAP to predict failure risk and explain which parameters contribute to the prediction.
- Hidden Pattern Detection: Uses Isolation Forest and K-Means clustering to identify anomalies and group similar failure behaviors.
- Natural-Language Copilot: Gemini allows engineers to ask questions such as "Why did this configuration fail?" and get insights directly from the analyzed data.



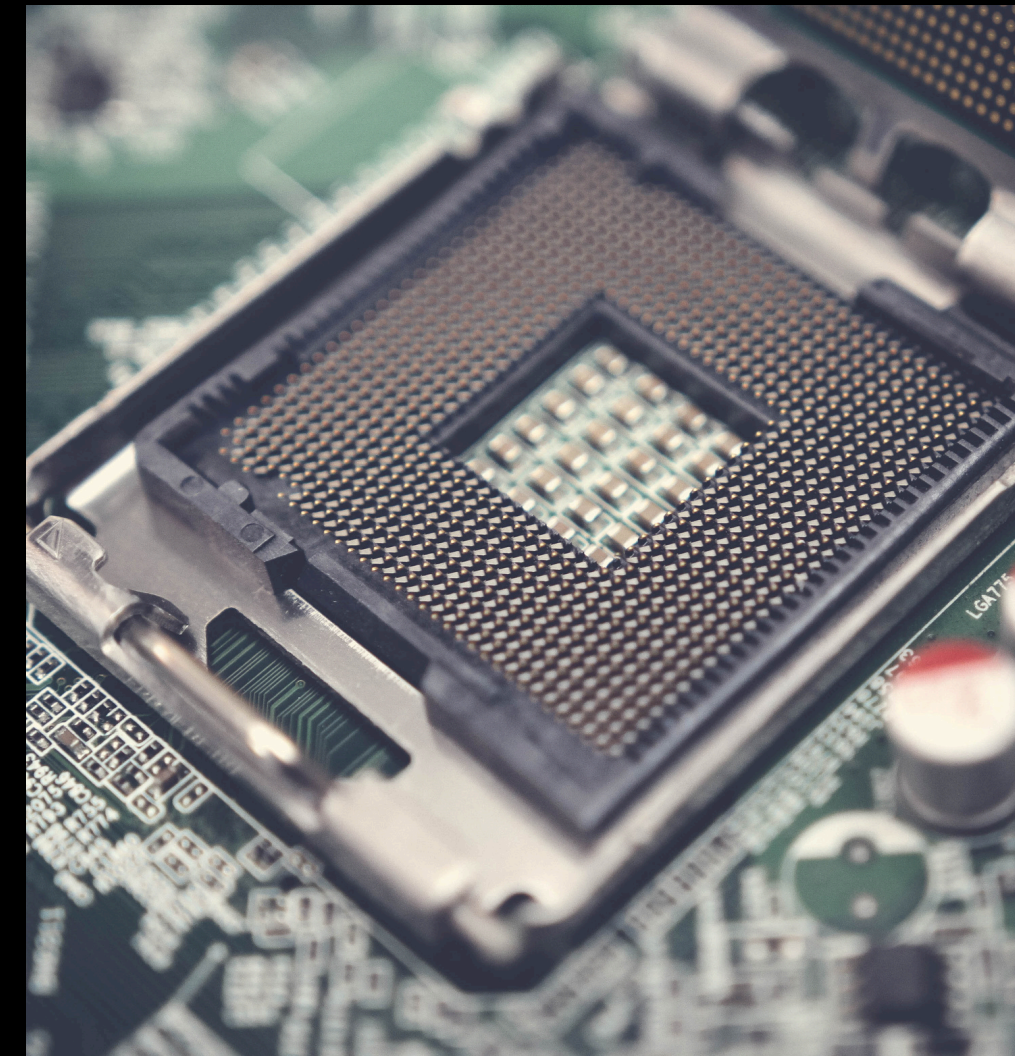
IMPACT:

- Reduces manual log-analysis effort when dealing with thousands of execution records.
- Speeds up debugging by highlighting important failure patterns and configuration differences.
- Enables proactive testing through failure-risk prediction and early-warning scores.
- Improves configuration selection by recommending configurations based on historical performance.
- Supports VLSI and semiconductor verification by turning large volumes of test data into actionable engineering insights.

FUTURE WORK:

- Real-time log analysis instead of only uploaded datasets.
- Scalable processing for millions of execution logs and larger verification environments.
- Advanced causal analysis to better distinguish correlation from actual failure causes.
- Smarter configuration optimization to automatically identify the best configuration for upcoming runs.
- Proactive AI Copilot that can automatically detect important issues and suggest debugging actions.
- Integration with existing VLSI verification workflows for continuous monitoring and analysis.

INDUSTRY CHALLENGES



01 SUPPLY CHAIN FRAGILITY
(SHORTAGES, DISRUPTIONS)

03 TALENT SHORTAGES IN
ENGINEERING AND MANUFACTURING

02 HIGH CAPITAL COSTS (MODERN FAB
COSTS CAN REACH TENS OF BILLIONS)

04 SUSTAINABILITY CONCERNS:
ENERGY AND WATER USE

THANK YOU
-TEAM NANOMETER